

## Final Pandas Lab-Practice Notebook

```
import pandas as pd
```

## ✓ 1. DataFrame from dictionary

```
data = {'calories':[420,380,390], 'duration':[50,40,45]}  
df = pd.DataFrame(data)  
print(df)
```

|   | calories | duration |
|---|----------|----------|
| 0 | 420      | 50       |
| 1 | 380      | 40       |
| 2 | 390      | 45       |

## ✓ 2. DataFrame example

```
mydataset = {'cars':['BMW','Volvo','Ford'], 'passings':[3,7,2]}  
myvar = pd.DataFrame(mydataset)  
print(myvar)
```

|   | cars  | passings |
|---|-------|----------|
| 0 | BMW   | 3        |
| 1 | Volvo | 7        |
| 2 | Ford  | 2        |

## ✓ 3. head, tail, info, describe

```
print(df.head())
```

|   | calories | duration |
|---|----------|----------|
| 0 | 420      | 50       |
| 1 | 380      | 40       |
| 2 | 390      | 45       |

```
print(df.tail())
```

|   | calories | duration |
|---|----------|----------|
| 0 | 420      | 50       |
| 1 | 380      | 40       |
| 2 | 390      | 45       |

```
#Summary of the DataFrame that includes the data types and non-null counts for each column.  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3 entries, 0 to 2  
Data columns (total 2 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   calories    3 non-null      int64  
1   duration    3 non-null      int64  
dtypes: int64(2)  
memory usage: 180.0 bytes
```

```
df.describe()
```

|              | calories   | duration |
|--------------|------------|----------|
| <b>count</b> | 3.000000   | 3.0      |
| <b>mean</b>  | 396.666667 | 45.0     |
| <b>std</b>   | 20.816660  | 5.0      |
| <b>min</b>   | 380.000000 | 40.0     |
| <b>25%</b>   | 385.000000 | 42.5     |
| <b>50%</b>   | 390.000000 | 45.0     |
| <b>75%</b>   | 405.000000 | 47.5     |
| <b>max</b>   | 420.000000 | 50.0     |

## ✓ 4. Column selection

```
print(df['calories'])
print(df[['calories', 'duration']])
```

```
0    420
1    380
2    390
Name: calories, dtype: int64
   calories  duration
0        420         50
1        380         40
2        390         45
```

```
# Identify missing values in the DataFrame.
print(df.isnull().sum())
```

```
calories    0
duration    0
```

```
dtype: int64
```

```
print(df.sum())
```

```
calories    1190
duration     135
dtype: int64
```

## ✓ 5. loc and iloc

```
print("loc:")
print(df.loc[0])
print(df.loc[0:2, ['calories', 'duration']])
```

```
print("\niloc:")
print(df.iloc[0])
print(df.iloc[1:3])
print(df.iloc[:, 0])
```

```
loc:
calories    420
duration     50
Name: 0, dtype: int64
   calories  duration
0         420         50
1         380         40
2         390         45
```

```
iloc:
calories    420
duration     50
Name: 0, dtype: int64
   calories  duration
1         380         40
2         390         45
0         420
1         380
```

```
2    390
Name: calories, dtype: int64
```

## ✓ 6. Series and labels

```
a = [1,7,2]
print(pd.Series(a))
print(pd.Series(a, index=['x','y','z']))
```

```
0    1
1    7
2    2
dtype: int64
x    1
y    7
z    2
dtype: int64
```

## ✓ 7. Unique values and Boolean filtering

```
people = pd.DataFrame({
    'Name': ['A', 'B', 'C', 'D', 'E'],
    'Age': [20, 21, 20, 23, 21],
    'Salary': [30000, 40000, 35000, 50000, 42000]
})
print(people['Age'].unique())
print(people[people['Age'] >= 21])
```

```
[20 21 23]
   Name  Age  Salary
1    B   21   40000
3    D   23   50000
4    E   21   42000
```

## ✓ 8. Read CSV

```
# After uploading data.csv to Colab:
# df_csv = pd.read_csv('data.csv')
# print(df_csv.head())
# print(df_csv.to_string())

# Demo CSV:
from io import StringIO
csv_data = StringIO('''Name,Age,Salary
A,20,30000
B,21,40000
C,22,35000
D,23,50000
''')
df_csv = pd.read_csv(csv_data)
print(df_csv.to_string())
```

|   | Name | Age | Salary |
|---|------|-----|--------|
| 0 | A    | 20  | 30000  |
| 1 | B    | 21  | 40000  |
| 2 | C    | 22  | 35000  |
| 3 | D    | 23  | 50000  |

## ✓ 9. Delete a column

```
df_drop = people.copy()
df_drop.drop(columns=['Salary'], inplace=True)
print(df_drop)
```

|   | Name | Age |
|---|------|-----|
| 0 | A    | 20  |
| 1 | B    | 21  |
| 2 | C    | 20  |

```
3   D   23  
4   E   21
```

## ✓ 10. Delete a row

```
df_drop = people.copy()  
df_drop.drop(1, axis=0, inplace=True)  
print(df_drop)
```

|   | Name | Age | Salary |
|---|------|-----|--------|
| 0 | A    | 20  | 30000  |
| 2 | C    | 20  | 35000  |
| 3 | D    | 23  | 50000  |
| 4 | E    | 21  | 42000  |

## Practice Questions

1. Create a Student DataFrame.
2. Use head(), tail(), info(), describe().
3. Select columns with loc and iloc.
4. Find unique values.
5. Filter rows using a Boolean condition.
6. Read a CSV.
7. Delete one column.
8. Delete one row.

